

VARUN KAPOOR

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PROFILE SUMMARY

Backend-focused Information Technology student with hands-on experience engineering scalable, production-grade systems in Java and Spring Boot. Demonstrated ability to architect geospatial data pipelines, integrate AI-driven APIs, and secure applications with JWT-based authentication. Passionate about performance optimization, clean system design, and solving complex problems through data-driven backend engineering.

TECHNICAL SKILLS

Languages: Java (Core & Advanced), C, C++, Dart, JavaScript, HTML/CSS, SQL

Frameworks: Spring Boot, Hibernate, Servlets, JSP, Flutter, RESTful APIs, JavaMail API

Databases: PostgreSQL, PostGIS, MySQL, Redis

Tools: Git, Docker, Postman, Leaflet.js, Mapbox, Linux

Concepts: OOP, Data Structures & Algorithms, DBMS, Computer Networks, Operating Systems

EDUCATION

B.Tech, Information Technology | Medi-Caps University, Indore | Expected 2027

12th Grade (CBSE) | English Medium | 2023

PROJECTS

Strideun — GPS Territory Painting Fitness App

Spring Boot, Java 21, PostgreSQL, PostGIS, Redis, Flutter, Mapbox, JWT

GitHub: github.com/Varunnnnn11/strideUN-Backend | github.com/Varunnnnn11/strideUN-Flutter

- Architected the full backend for a gamified real-world fitness platform where users claim map territories by physically traversing GPS zones, supporting concurrent multi-user ownership via PostGIS spatial indexing on PostgreSQL.
- Reduced GPS processing overhead by ~40% by engineering a 30-second coordinate batching pipeline combined with a 6-layer Kalman filter algorithm, eliminating hardware noise and blocking GPS spoofing before data reaches the database.
- Eliminated redundant leaderboard database queries by integrating Redis caching, enabling sub-millisecond cache-hit responses on live rankings at scale.
- Hardened all API endpoints by implementing stateless JWT authentication with rotating access and refresh token pairs, enforcing session integrity across Flutter clients and the Spring Boot backend.
- Automated database lifecycle management via Spring Scheduler jobs — sustaining PostgreSQL connection keep-alives and executing zone-decay cleanup routines to prevent stale GPS data accumulation.

EzEv — Smart EV Charging & AI Route Planning Platform

Java, Servlets, JSP, Hibernate, Leaflet.js, Groq AI (Llama 3), OpenChargeMap API, JavaMail API

GitHub: github.com/Varunnnnn11/EZEV

- Engineered a full-stack web platform enabling EV owners and hosts to list, discover, and book charging stations, managing end-to-end booking state through a normalized relational schema in MySQL via Hibernate ORM.
- Designed an AI-powered waypoint algorithm combining the Groq API (Llama 3), the Haversine formula, and real-time battery/range inputs to recommend optimal charging stops, reducing route-planning complexity for drivers.
- Integrated live data from the OpenChargeMap API into an interactive Leaflet.js map, surfacing real-time station availability, connector types, and pricing without page reloads.
- Automated transactional email confirmations via JavaMail API, delivering real-time receipts and wallet summaries upon payment to increase booking completion confidence.
- Modeled complex many-to-many relationships between Users, Stations, and Bookings using Hibernate ORM annotations, achieving scalable, schema-agnostic persistence and eliminating raw SQL coupling.

CERTIFICATIONS

- **Java 2.0 (Core & Advanced) Training** — Universal Informatics, Indore (Feb 2025 – May 2025). Hands-on training program covering Core Java, OOP, and Advanced Java concepts.
- **Spring & Spring Boot (J2EE Enterprise Edition) Training** — Universal Informatics, Indore (Sept 2025 – Nov 2025). Industrial training covering Spring Framework, Spring Boot, REST APIs, Hibernate ORM, and enterprise application development.
- **Python 3.4.3 Training** — EduPyramids, SINE, IIT Bombay (Oct 2025). Completed Spoken Tutorial Project examination conducted remotely from IIT Bombay; Credits: 4.
- **Linux Training** — Spoken Tutorial Project, IIT Bombay (Mar 2025). Completed examination conducted remotely from IIT Bombay; Credits: 2.